Week 2 Module:

Date: 17/06/2026

1. Find Student Roll Number (Linear Search)

Problem Statement

Your class teacher has a list of student roll numbers stored in the order they registered. Since the list is unsorted, you need to find whether a particular student roll number exists.

Write a program to search for the given roll number using **Linear Search**.

Input Format

- First line contains integer N (number of students)
- Second line contains N roll numbers
- Third line contains integer X (roll number to search)

Output Format

- Print "Found at index i" if present
- Otherwise print "Not Found"

Constraints

- $1 \le N \le 1000$

Example

Input

5
101 105 113 110 130
110

Output

Found at index 3

```
# include <stdio.h>
# include <conio.h>

void main
{

  int a[100], n, x, i, found=0;

   clrscr();

  printf(" enter no. of students:");
  scanf("%d", &n);

  printf(" enter roll no..:\n");
   for(i=0; i<n; i++)
  {
     scanf("%d", &a[i]);

  }

   printf("enter roll no. to search:
          ");

   scanf("%d", &x);
```

```
for (i=0; i<n; i++)
{
  if (a[i] == x)
  {
    printf(" found at index %d", i);
    found = 1;
    break;
  }
}
if (found == 0)
{
  printf("Not found");
}
getch();
}
```

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ADHAYA.CPP

```c
#include<stdio.h>
#include<conio.h>

void main()
{
  int a[100],n,x,i,found=0;

  clrscr();

  printf("enter number of students:");
  scanf("%d",&n);

  printf("enter roll number:\n");
  for(i=0;i<n;i++)
  {
    scanf("%d",&a[i]);
  }

  printf("enter roll number to search:");

  scanf("%d",&x);
```

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ADHAYA.CPP

```
scanf("%d",&x);
for(i=0;i<n;i++)
{
   if(a[i]==x)
    {
      printf("found at index %d",i);

      found=1;
      break;
    }
  }
    if (found==0)
  {
   printf("not found");

    }

  getch();
}
```

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```
enter number of students:5
enter roll number:
101 105 113 110 130
enter roll number to search:110
found at index 3_
```

2. Search Product ID (Binary Search)

Problem Statement

An online store keeps product IDs sorted in ascending order. A customer wants to know whether a particular product exists in inventory.

Write a program to search the product using **Binary Search**.

Input Format

- First line contains N
- Second line contains sorted product IDs
- Third line contains target ID

Output Format

Print:

Product Available

or

Product Not Available

Constraints

- 1 ≤ N ≤ 10^5

Example

Input

6
10 20 30 40 50 60
40

Output

Product Available

```
# include <stdio.h>
# include <conio.h>

void main ()
{

int a[100], n, x;
int low, high, mid;

clrscr();

printf(" enter no. of products : ");
scanf("%d", &n);

printf("enter sorted products IDs :\n
for (int =0; i<n; i++)
{
scanf("%d". &a[i]);
}

printf ("enter product ID to
       search:");
scanf("%d", &x);

low = 0;
high = n-1;
```

```
while (low <= high)
{
  mid (low+ high)/2;
  if (a[mid] = = x)
  {
   printf ("product available");

   getch();
   return;
  }
  else if (a[mid] <x)
  {
    low = mid +1;
  }
  else
  {
    high = mid - 1;
  }
}

printf ("product Not available");
```

```
getch();
}
```

ADHAYA.CPP

```c
#include<stdio.h>
#include<conio.h>

void main()
{
   int a[100],n,x;
   int low,high,mid;

   clrscr();

   printf("enter number of products:");
   scanf("%d",&n);

   printf("enter sorted product ids:\n");
   for(int i=0;i<n;i++)
   {
     scanf("%d",&a[i]);
   }

   printf("enter product id to search:");
```

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ADHAYA.CPP

```
scanf("%d",&x);
low=0;
high=n-1;

while(low<=high)
{
  mid=(low+high)/2;
  if(a[mid]==x)
  {
     printf("product available");
     getch();
     return;

   }
   else if(a[mid]<x)
   {
     low=mid+1;
   }
   else
   {
```

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[■] ADHAYA.CPP

```
        {
          high=mid-1;
        }
    }
        printf("product not available");

      getch();
  }
```

```
enter number of products:6
enter sorted product ids:
10 20 30 40 50 60
enter product id to search:40
product available
```

3. Arrange Exam Scores (Bubble Sort)

Problem Statement

A teacher wants to arrange students' exam scores in ascending order to prepare the result sheet.

Write a program using **Bubble Sort**.

Input Format

- First line contains N
- Second line contains N integers

Output Format

Print the sorted array.

Example

Input

5
89 45 78 12 99

Output

12 45 78 89 99

```c
# include <stdio.h>
# include <conio.h>

void main()
{
 int a[100];
 int n, i, j, temp;

 clrscr();

 printf("enter number of students:");
 scanf("%d", &n);

 printf("enter exam scores:\n");
 for (i=0; i<n; i++)
 {
  scanf("%d", &a[i]);
 }
```

```
for (i = 0; i<n-1; i++)
{
  for (j = 0; j<n-1-i; j++)
  {
    if (a[j] > a[j+1])
    {

      temp = a[j];
      a[j] = a[j+1];
      a[j+1] = temp;
    }
  }
}

printf("scores in ascending
order: \n");

for(i=0; i<n; i++)
{
printf("%d", a[i]);
```

}

getch();

}

ADHAYA.CPP

```
#include<stdio.h>
#include<conio.h>

void main()
{
  int a[100];
  int n,i,j,temp;

  clrscr();

  printf("enter number of students:");
  scanf("%d",&n);

  printf("enter exam scores:\n");
  for( i=0;i<n;i++)
  {
    scanf("%d",&a[i]);
  }
   for(i=0;i<n-1;i++)
   {
     for(j=0;j<n-1;j++)
```

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Compile Debug Project

ADHAYA.CPP

```
    for(j=0;j<n-1;j++)
     {
       if(a[j]>a[j+1])
       {
          temp=a[j];
          a[j]=a[j+1];
          a[j+1]=temp;
       }
     }
  }
    printf("scores in ascending order:\n");
     for(i=0;i<n;i++)
     {
       printf("%d ",a[i]);
     }
   getch();
 }
```

```
enter number of students:5
enter exam scores:
89 45 78 12 99
scores in ascending order:
12 45 78 89 99 _
```

4. Arrange Library Book Numbers (Insertion Sort)

Problem Statement

A librarian receives book serial numbers one by one and wants to place each new book in the correct position.

Write a program using **Insertion Sort** to arrange book numbers.

Input Format

- First line contains N
- Second line contains N integers

Output Format

Print sorted book numbers.

Example

Input

5
55 23 87 11 34

Output

11 23 34 55 87

```
# include <stdio.h>
# include <conio.h>

Void main()
{

int a[100];
int n, i, j, key;

clrscr();

printf("enter no. of books: ");
scanf("%d", &n);

printf("enter books no: \n");
for (i=0; i<n; i++)
{
  scanf("%d", &a[i]);
}
for (i=1; i<n; i++)
{
  key = a[i];
  j = i-1;
```

```
while (j>=0 && a[j]>key)
{
  a[j+1]= a[j];
  j--;
}
  a[j+1] = key;
}

printf("sorted books no : \n");
for(i=0; i<n; i++)
 {
  printf("%d", a[i]);
 }

  getch();

}
```

```c
#include<stdio.h>
#include<conio.h>

void main()
{
   int a[100];
   int n,i,j,key;

   clrscr();

   printf("enter number of books:");
   scanf("%d",&n);

   printf("enter books numbers:\n");
   for( i=0;i<n;i++)
   {
     scanf("%d",&a[i]);
   }
    for(i=1;i<n;i++)
    {
      key=a[i];
```

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ADHAYA.CPP

```
    key=a[i];
    j=i-1;

    while(j>=0 && a[j]> key);
    {
      a[j+1]=a[j];
      j--;
    }
     a[j+1]=key;
  }
    printf("sorted books numbers:\n");
     for(i=0;i<n;i++)
     {
       printf("%d ",a[i]);
     }
   getch();
}
```

```
enter number of books:5
enter books numbers:
55 23 87 11 34
sorted books numbers:
11 23 34 55 87 _
```

5. Rank Players by Score (Selection Sort)

Problem Statement

A gaming club wants to rank players based on scores from lowest to highest.

Write a program using **Selection Sort**.

Input Format

- First line contains N
- Second line contains player scores

Output Format

Print sorted scores.

Example

Input

5
70 20 90 10 40

Output

10 20 40 70 90

```
# include <stdio.h>
# include <conio.h>

void main()
{
int a[100], i, n, j, min, temp;

clrscr();

printf("enter no. of players:");
scanf("%d", &n);

printf("enter players scores:\n");
for(i=0; i<n; i++)
{
scanf("%d", &a[i]);
}
for(i=0; i<n-1; i++)
{
min=i;
for(j=i+1; j<n; j++)
```

```
{
    if (a[j] < a(min])
    {
        min = j;
    }
}
    temp = a[i];
    a[i] = a[min];
    a[min] = temp;
}
    printf ("\n sorted scores: \n");
    for (i=0; i<n; i++).
    {
        printf ("%d", a[i]);
    }

    getch();
}
```

ADHAYA.CPP

```
#include<stdio.h>
#include<conio.h>

void main()
{
   int a[100];
   int n,i,j,key;

   clrscr();

   printf("enter number of books:");
   scanf("%d",&n);

   printf("enter books numbers:\n");
   for( i=0;i<n;i++)
   {
     scanf("%d",&a[i]);
   }
    for(i=1;i<n;i++)
    {
      key=a[i];
```

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ADHAYA.CPP

```
key=a[i];
j=i-1;

while(j>=0 && a[j]> key)
{
  a[j+1]=a[j];
  j--;
}
 a[j+1]=key;
}
printf("sorted books numbers:\n");
 for(i=0;i<n;i++)
 {
  printf("%d ",a[i]);
 }
getch();
}
```

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```
enter number of players:5
enter players scores:
70 20 90 10 40

sorted scores:
10 20 40 70 90 _
```